

Fine Grained Pet Breed Classification

A deep learning case study on fine-grained cat and dog breed recognition

37
breeds

25,779
images

EfficientNet-Bo
90.46%
test accuracy



The problem was finer than ordinary classification

Distinguishing breeds requires noticing subtle visual cues that look almost the same.

37 breeds

with subtle visual differences

12

cat breeds

25

dog breeds

18,045

train images

3,867

validation images

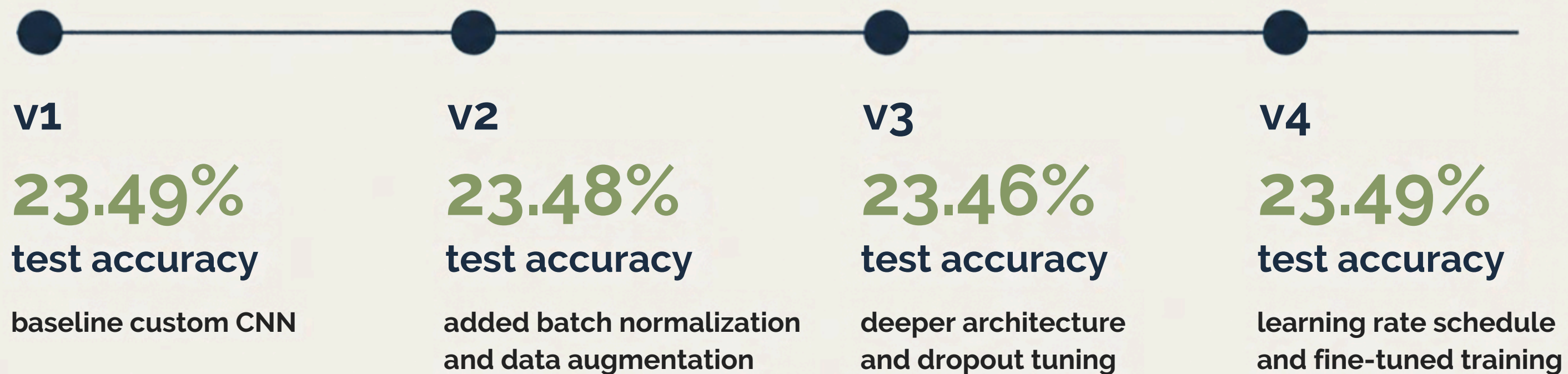
3,867

test images



The first model was not enough

Iterating our custom CNN improved results, but we reached its performance ceiling.

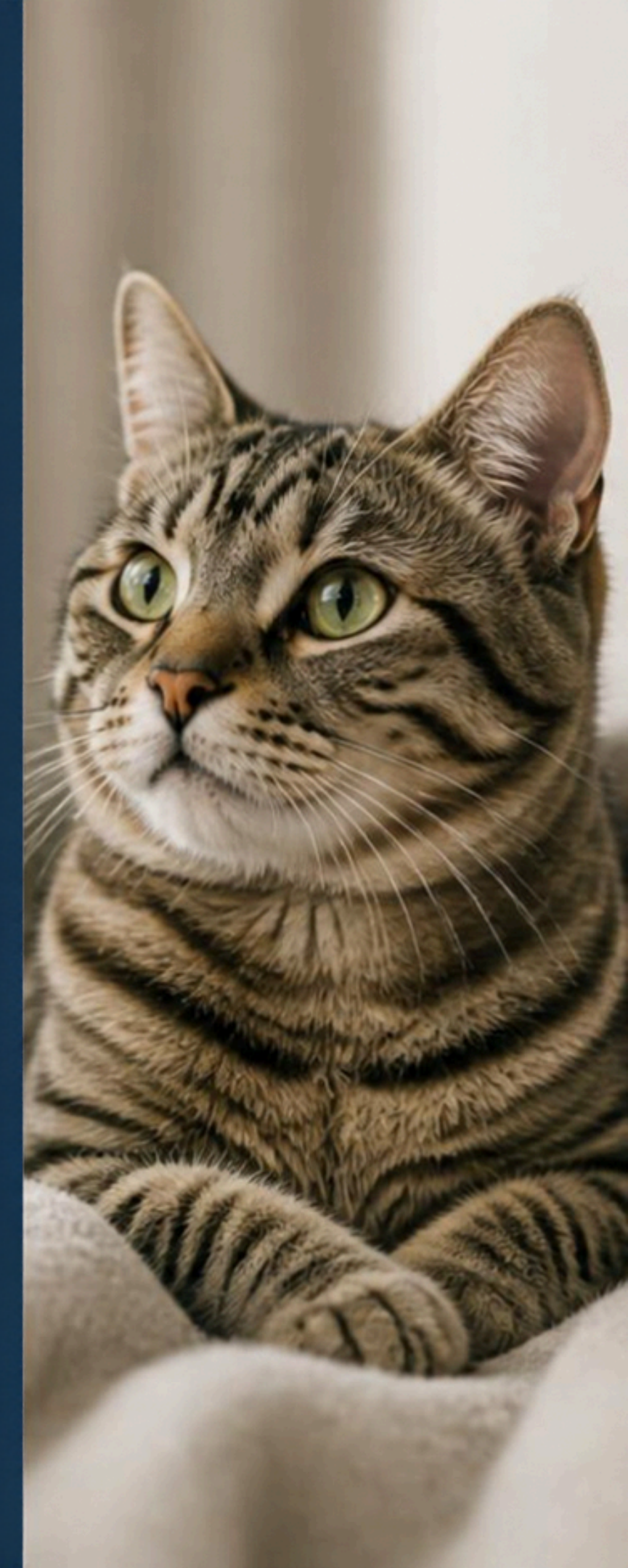


The experiments revealed the **ceiling of the custom CNN**.



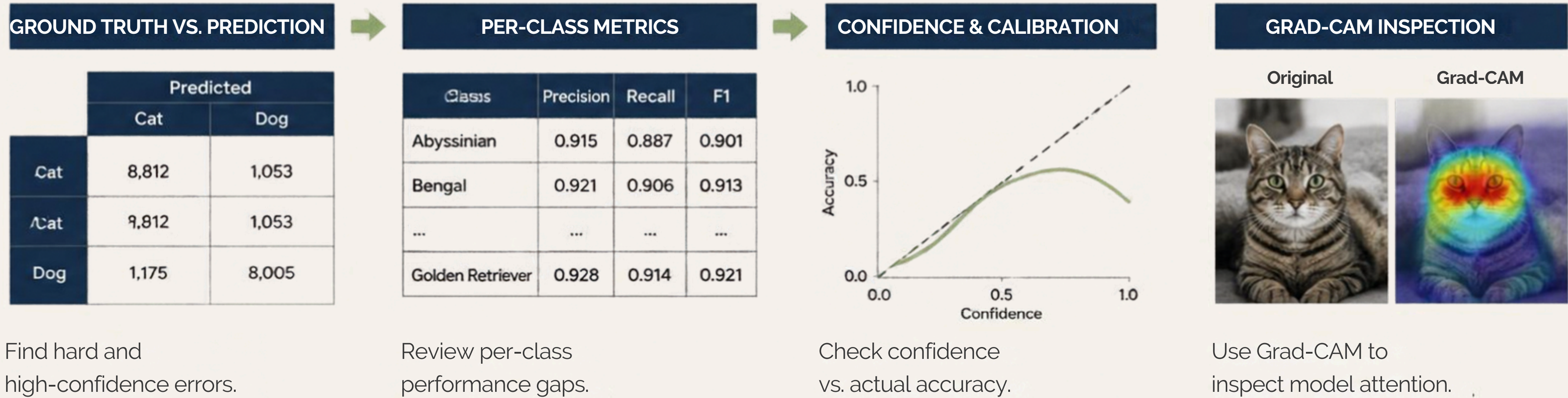
Transfer learning changed the ceiling

Custom CNN v4		ResNet18		EfficientNet-Bo			
23.49%		88.63%		90.46%			
top-1 accuracy		top-1 accuracy		top-1 accuracy			
parameters		11.20M		4.05M			
parameters		parameters		parameters			
Adam		1e-4		10 epochs		CUDA	
optimizer		learning rate		training		hardware	



Accuracy was only the beginning

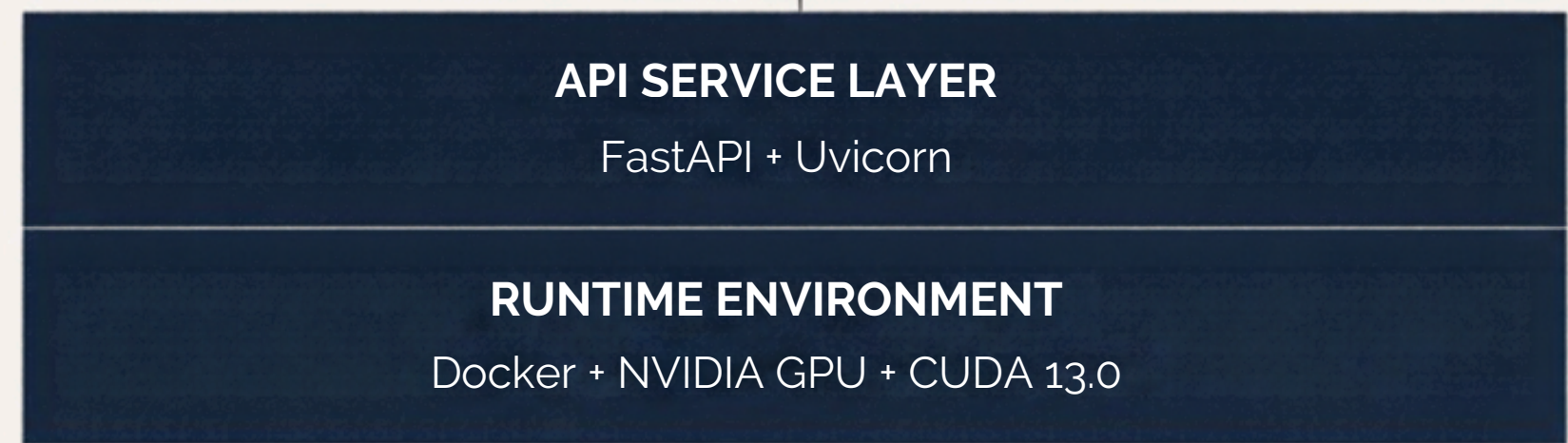
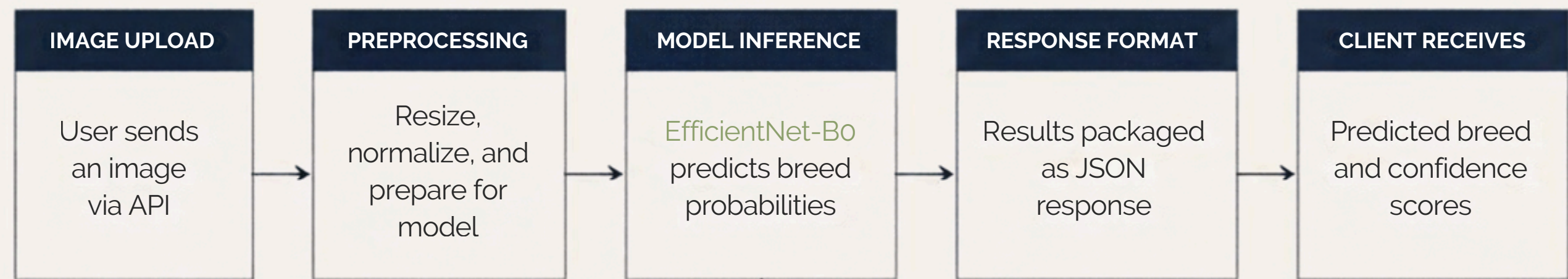
Evaluation as a diagnostic workflow, not a dashboard.



0.9021

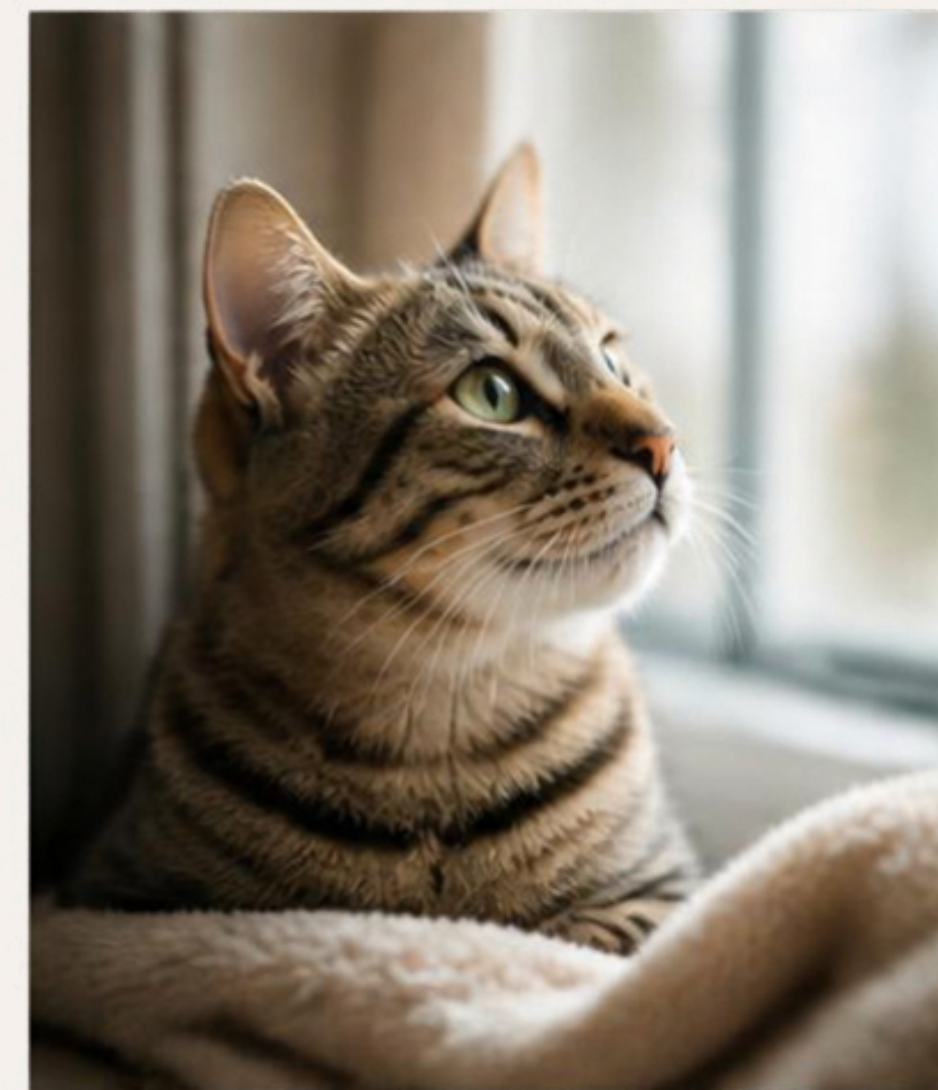
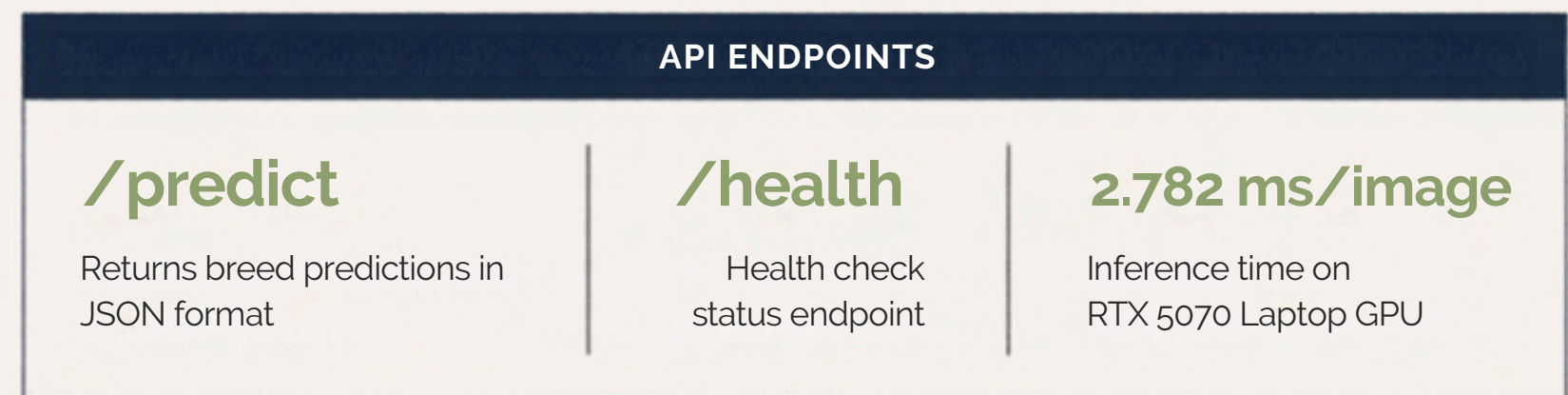
Macro F1

From notebook to production



2.782
ms/image

on RTX 5070
Laptop GPU



A production-ready pipeline built for speed and scale.

Reliable predictions with industry-standard deployment.